Name: _____

Astronomy 4

Date:

Time:

Total marks available: 37

Total marks achieved: _____

Q1.

Satellites are used to gather data about the origin of the Universe.

The Big Bang theory is a theory about the origin of the Universe.

Evidence for the Big Bang theory is provided by red-shift and CMB radiation.

(i) Describe what is meant by red-shift.

(2)

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(ii) Explain how red-shift provides evidence for the Big Bang theory.

(2)

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(iii) The Cosmic Background Explorer (COBE) satellite observed CMB radiation from 1989 to 1993.
State what the 'M' in CMB radiation stands for.

(1)

.....

(iv) State what is meant by 'cosmic background radiation'.

(1)

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(v) Explain how the presence of CMB radiation provides evidence for the Big Bang theory.

(2)

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(Total for question = 8 marks)

Q2.

Figure 7 shows two hydrogen spectra.

One spectrum is taken from a source on Earth.

The other spectrum is taken from a source on a distant galaxy.

The spectral lines in the spectrum from the distant galaxy are shifted towards the red end of the spectrum.

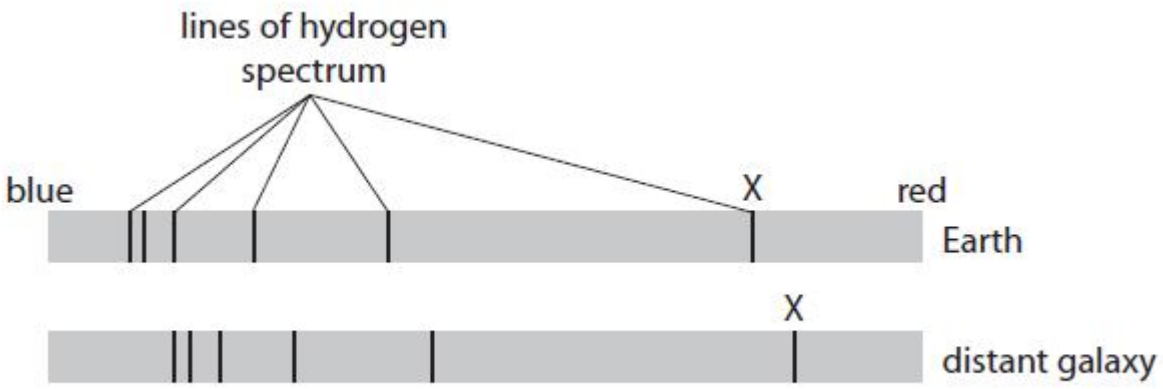

Figure 7

The wavelength of line X on Earth, λ_o = 6.56 x 10^{-7}m

The wavelength of line X from the distant galaxy, λ_g = 6.72 x 10^{-7}m

The red shift (z) is given by the equation

$$z = \frac{(\lambda_g - \lambda_o)}{\lambda_o}$$

(i) Show that the red shift for the light from the distant galaxy is about 0.025

(2)

(ii) The galaxy is moving away from the Earth at velocity, v (recession velocity).

The velocity of light, c = 3.00 x 10^8 m/s
The recession velocity is given by the equation.

$$v = z \times c$$

Calculate the recession velocity of the distant galaxy.

(2)

recession velocity = ... m/s

(iii) The wavelength of the spectral line X measured for a more distant galaxy was 6.92×10^{-7} m.

Explain how this provides evidence that the Universe is expanding.

(2)

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.....

(Total for question = 6 marks)

Q3.

Our Sun is a main sequence star.
It will stay as a main sequence star for millions of years, continuing to radiate energy.

(i) Our Sun was formed from a nebula.

Describe the formation of our Sun from nebula to main sequence.

(2)

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.....

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.....

(ii) Our Sun will go through many stages.

Which of these stages will our Sun go through?
Put a cross (☒) in the box next to your answer.

(1)

☐ **A** a black hole

☐ **B** a neutron star

☐ **C** a supernova

☐ **D** a white dwarf

Q4.

CMB radiation provides evidence that the Universe had a definite beginning.

Use the table in Figure 7 to give a typical value for the wavelength of CMB radiation.

type of radiation	typical wavelength
gamma rays	1.0×10^{-12} m
X-rays	3.0×10^{-11} m
ultraviolet	200 nm
visible	600 nm
infrared	4.0 μm
microwaves	1.0 mm
radio waves	50 m

Figure 7

(2)

wavelength = ...

(Total for question = 2 marks)

Q5.

The table in Figure 16 shows some data about four stars.

star	mass in kg	age in years
Antares	2.6×10^{31}	1.5×10^{7}
Arcturus	2.2×10^{30}	7.1×10^{9}
Proxima Centauri	2.4×10^{29}	4.8×10^{9}
The Sun	2.0×10^{30}	4.6×10^{9}

Figure 16

(i) State which star in the table in Figure 16 is **most likely** to become a Black Hole.

(1)

.....

(ii) Using data from the table in Figure 16, give a reason for your answer to part (i).

(1)

.....

.....

(Total for question = 2 marks)

Q6.

Figure 3 shows a photograph of galaxy D.

This photograph was taken by a student at his home.

(Source: Paul Curtis)

Figure 3

State **two** ways that the student can improve the observational techniques so that the quality of the image is improved.

(2)

1 .....

.....

2 .....

.....

(Total for question = 2 marks)

Q7.

During the evolution of a star, the nebula collapses and becomes a main sequence star.

(i) State what causes the nebula to collapse.

(1)

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(ii) Explain why the nebula stops collapsing as it becomes a main sequence star.

(3)

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(Total for question = 4 marks)

Q8.

Two theories about the Universe are the Steady State Theory and The Big Bang Theory.

(i) The table shows some ideas about the Universe.

Which row of the table applies to the Steady State Theory?
Put a cross (☒) in the box next to your answer.

(1)

	the Universe.....	the Universe.....
☐ A	... is not expanding	... had a beginning
☐ B	... is expanding	... had a beginning
☐ C	... is not expanding	... did not have a beginning
☐ D	... is expanding	... did not have a beginning

(ii) State the name of the discovery that persuaded most scientists to prefer the Big Bang Theory to the Steady State Theory.

(1)

.....

Q9.

Figure 2 shows some lines in the absorption spectra from four different galaxies (A, B, C, and D) and from a laboratory source.

All the spectra are aligned and to the same scale.

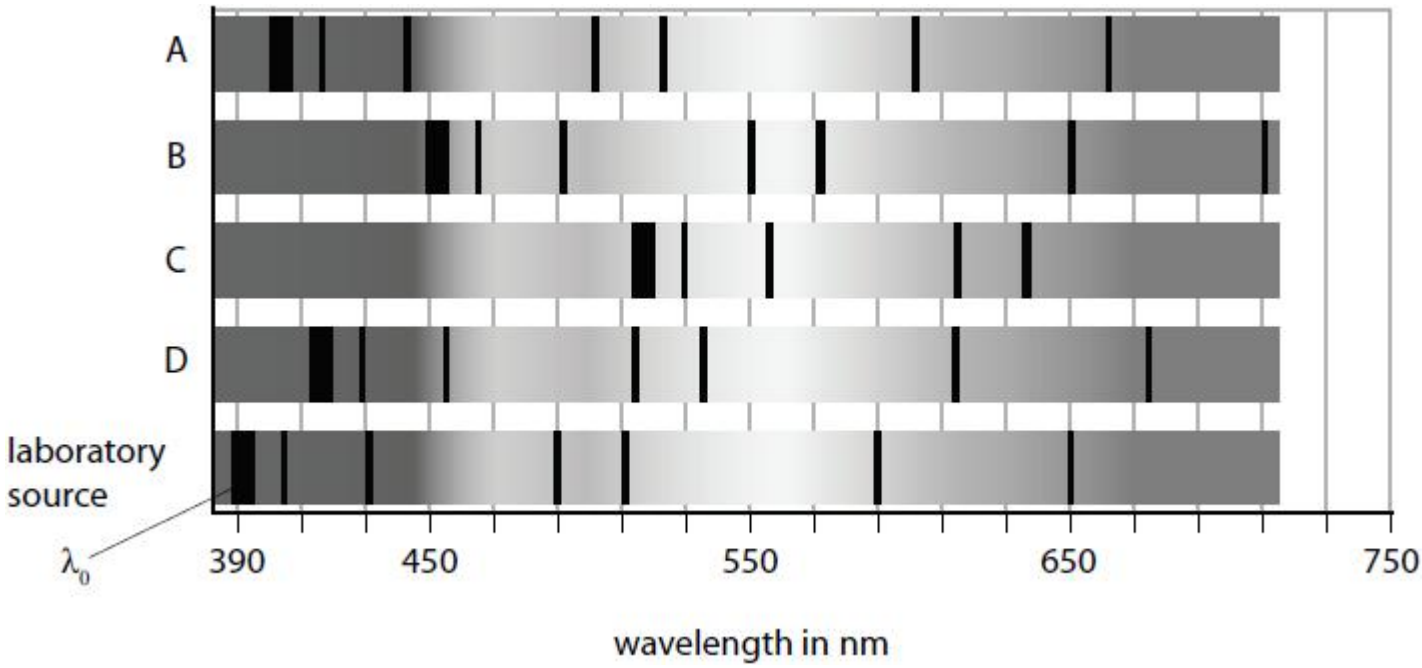

Figure 2

(i) Explain, using Figure 2, which galaxy is furthest away from us.

(3)

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(ii) In Figure 2, the reference wavelength, λ_0, is shown at 390 nm.

Estimate the change in the reference wavelength, $\Delta\lambda$, for the light from galaxy D.

(1)

$\Delta\lambda$ = ... nm

(iii) Calculate the speed, *v*, of galaxy D.

Use the equation

$$v = c\frac{\Delta\lambda}{\lambda_0}$$

[c = speed of light = 3×10^8 m/s]

(2)

v = ... m/s

(Total for question = 6 marks)

Q10.

A star has evolved to become a neutron star.

The mass, M, of the neutron star, of radius R, is given by

$$M = \frac{4 \times \pi \times D \times R^3}{3}$$ where D is a constant

$M = 4 \times 10^{30}$ kg
$D = 6 \times 10^{17}$ kg/m^3
Use the equation to calculate the value for R.

(2)

R = ... m

(Total for question = 2 marks)